

Cloud Architecture Specifications 2025

1. Executive Summary and SLA Targets

The NextGen Cloud Platform provides enterprise-grade infrastructure with guaranteed availability of 99.999% uptime across all Tier-4 availability zones. The system enforces a strict p99 latency SLA limit of 15ms for internal RPC calls. In the event of an unplanned primary region outage, the automated disaster recovery protocol executes cross-region failover within 30 seconds with zero committed data loss.

2. Availability Zone Topology

Deployments span a minimum of three distinct physical availability zones separated by at least 25 kilometers to protect against local environmental disasters. Inter-zone network traffic travels over dedicated dark fiber with redundant 400 Gbps optical transport.

3. Distributed Storage Layer and Consensus

The distributed object and metadata store relies on the Raft consensus algorithm. Each cluster maintains a quorum of 5 replicas, requiring a minimum of 3 positive acknowledgments before committing any write to disk. Write-ahead logs (WAL) are flushed to non-volatile NVMe storage synchronously before ack. Snapshot compaction runs automatically every 6 hours or whenever the WAL exceeds 10,000 log entries.

4. Storage Durability and Replication

Object data achieves an annual durability guarantee of eleven nines (99.999999999%). Data chunks are erasure coded using an 8+4 Reed-Solomon scheme across failure domains. Storage tiering transitions cold objects to deep archive storage after 90 days of inactivity.

5. Security, Encryption, and Egress Pricing

All data at rest is encrypted using AES-256-GCM with customer-managed keys rotated annually. In-transit communication requires mandatory TLS 1.3 encryption with forward secrecy. Tenant API access is strictly throttled to a default rate limit of 10,000 req/sec per tenant. Standard external data egress pricing is fixed at \$0.05 per gigabyte for the first 50 terabytes, discounted to \$0.03 per gigabyte thereafter.

6. Audit Logging and Compliance

Immutable audit trails capture all administrative operations and access requests. Audit logs are retained for 7 years in write-once-read-many (WORM) compliant storage fulfilling SOC-2 Type II, ISO 27001, and HIPAA compliance mandates.